SCOTT A. STONE

scott.allan.stone [at] gmail.com

EDUCATION

University of Alberta
Doctor of Philosophy (Psychology) 2023
Supervisors: Dr. Craig S. Chapman & Dr. Anthony Singhal
Thesis: *Eye and body tracking in the lab, in the wild, and in the clinic*

University of Lethbridge
Master of Science (Neuroscience) 2017
Supervisors: Dr. Matthew S. Tata & Dr. Claudia L.R. Gonzalez
Thesis: *Using Auditory Augmented Reality to Understand Visual Scenes*

University of Lethbridge
Bachelor of Science (Neuroscience) 2015
Supervisor: Dr. Claudia L.R. Gonzalez

RESEARCH EXPERIENCE

The Action in Complex Environments Lab, University of Alberta
Graduate Student (Ph.D) Sept. 2017 - May 2023
Supervisor: Dr. Craig S. Chapman

The Brain in Action Lab, University of Lethbridge
Graduate Student (M.Sc) & Research Assistant (B.Sc) Sept. 2013 - Aug. 2017
Supervisor: Dr. Claudia L.R. Gonzalez

Tata Lab, University of Lethbridge
Graduate Student (M.Sc) Sept. 2015 - Aug. 2017
Supervisor: Dr. Matthew S. Tata

Speech Neurogenetics Laboratory, University of Alberta
Summer Research Intern (UARE) May 2014 - Aug. 2014
Supervisors: Dr. Deryk S. Beal & Dr. Ivor Cribben

WORK EXPERIENCE

Reverb Robotics, Lethbridge
Co-founder May 2023 - current
Mitacs-funded postdoctoral researcher

University of Lethbridge, Lethbridge
Postdoctoral research fellow in the Brain in Action laboratory May 2023 - May 2025
Mitacs-funded postdoctoral researcher

BioWare, Edmonton
Mitacs Accelerate Intern - User Experience Researcher Feb. 2020 - Dec. 2022
Software development for analysis of hand and gaze patterns during user interface interaction

PUBLICATIONS

**equal contribution*

8. Ramirez, D. E. A., Grasse, L., **Stone, S.A.**, Tata, M.S., Gonzalez, C. L. (2025). AI-powered speech device as a tool for neuropsychological assessment of an older adult population: A preliminary study. *Acta Psychologica*. 257, 105084 — data

7. **Stone S.A.**, Chapman C.S. (2023). Unconscious frustration: dynamically assessing user experience using eye and mouse tracking. *ACM Symposium on Eye Tracking Research and Applications (ETRA); published in Proceedings of the ACM on Human-Computer Interaction.* 10.1145/3591137 — data

6. **Stone S.A.**, Boser Q.A., Dawson T.R, Vette A.H., Hebert J.S., Pilarski P.M., Chapman C.S. (2022). Generating accurate 3D gaze vectors using synchronized eye tracking and motion capture. *Behavior Research Methods.* 10.3758/s13428-022-01958-6 — data

5. **Stone S.A.**, Boser Q.A., Dawson T.R, Vette A.H., Hebert J.S., Pilarski P.M., Chapman C.S. (2021). Sub-centimeter 3D gaze vector accuracy on real-world tasks: an investigation of eye and motion capture calibration routines. *ACM Symposium on Eye Tracking Research and Applications (ETRA).* 10.1145/3450341.3458880

4. Wispinski N.J., **Stone S.A.**, Bertrand J.K., Ouellette Zuk A.A., Lavoie E.B., Gallivan J.P., Chapman C.S. (2021). Reaching for known unknowns: Rapid reach decisions accurately reflect the future state of dynamic probabilistic information. *Cortex.* 10.1016/j.cortex.2021.02.010 — data

3. **Stone S.A.**, Baker J., Olsen R., Gibb R., Doan J., Hoetmer J., Gonzalez C.L.R. (2019). Visual field advantage: Redefined by training? *Frontiers in Psychology.* 9:2764. 10.3389/fpsyg.2018.02764 — data

2. *Lavoie E.B., *Bertrand J.K., ***Stone S.A.**, *Wispinski N.J., *Sawalha J., *Chapman C.S. (2018). Examining the "species" of situated cognition in humans. Comment on "Cognition beyond representation: Varieties of situated cognition in animals". *Comparative Cognition & Behavior Reviews*, *13.* 10.3819/ccbr.2017.120003

1. **Stone S.A.**, Tata M.S. (2017). Rendering visual events as sounds: Spatial attention capture by auditory augmented reality. *PLOS ONE.* 12.8. e0182635. 10.1371/journal.pone.0182635 — data

CONFERENCE PRESENTATIONS

25. **Stone S.A.**, Aguilar Ramirez D.E., Grasse L., Tata M.S., Gonzalez C.L.R (2024). Assessing an AI-powered speech device as a tool for neuropsychological assessment of an older adult population. Presented at the CSBBCS CAPnet satellite symposium 2024. June 25; Edmonton, Alberta.

24. **Stone S.A.**, Chapman C.S., (2023). Unconscious frustration: dynamically assessing user experience using eye and mouse tracking. Talk presented at ACM Symposium on Eye Tracking Research and Applications (ETRA) 2023. May 30–June 2; Tübingen, Germany.

23. Wispinski N.J., **Stone S.A.**, Singhal A., Pilarski P.M., Chapman C.S., (2022). Primate-like perceptual decision making through deep recurrent reinforcement learning. Talk presented at Society for Neuroscience (SfN) 2022. Nov 12; San Diego, CA, USA.

22. **Stone S.A.**, Chapman C.S. (2022). Assessment of remote eye tracking using a video game menu navigation task. Eye and body movements during object interactions in real and virtual worlds. Poster presented at Society for Neuroscience (SfN) 2022. Nov 12; San Diego, CA, USA.

21. **Stone S.A.**, Boser Q.A., Dawson T.R, Vette A.H., Hebert J.S., Pilarski P.M., Chapman C.S. (2021). Sub-centimeter 3D gaze vector accuracy on real-world tasks: an investigation of eye and motion capture calibration routines. Talk presented at ACM Symposium on Eye Tracking Research and Applications. May 27; Reno, NV, USA. Virtual.

20. Beausoleil K., Jafferjee, T., Wispinski, N., **Stone S.A.**, Chapman, C.S. (2019). The Nature of Decision-Making: Human Behaviour vs Machine Learning. Poster presented at the WISEST poster presentation ceremony. August 15; Edmonton, AB.

19. Lavoie E.B., **Stone S.A.**, Hebert J.S., Chapman C.S. (2018). Eye and body movements during object interactions in real and virtual worlds. Poster presented at Society for Neuroscience (SfN). November 3; San Diego, CA, USA.

18. **Stone S.A.**, Lavoie E.B., Chapman C.S. (2018). Assessing different 3D gaze vector calibration techniques on synchronized motion capture and eye-tracking data. Poster presented at Society for Neuroscience (SfN). November 3; San Diego, CA, USA.

17. **Stone S.A.**, Dawson T.R., Boser Q., Hebert J.S., Chapman C.S. (2018). Using Lab Streaming Layer to collect synchronized multimodal datasets. Presented at the Canadian Action and Perception Network (CAPnet) Canadian Physiological Society (CPS) Satellite Symposium at the Canadian Association for Neuroscience (CAN) Conference. Vancouver BC.

16. Lee, V., **Stone S.A.**, Tata M.S. (2018). How many is too many at a cocktail party? Testing the limits of spatial resolution in complex auditory scenes. Presented at the Canadian Action and Perception Network (CAPnet) Canadian Physiological Society (CPS) Satellite Symposium at the Canadian Association for Neuroscience (CAN) Conference. Vancouver BC.

15. **Stone S.A.**, Gonzalez C.L.R., Tata M.S. (2017). Hearing What You See: A Novel Visual-to-Auditory Augmented Reality System. Presented at the Banff Annual Seminar in Cognitive Science (BASICS). May 5-6; Banff, AB.

14. Lee V., **Stone S.A.**, Tata M.S. (2017). Resolution of multiple talkers in a "cocktail party" depends on head movements. Presented at the Undergraduate Research in Science Conference of Alberta. April 7-8; Calgary AB.

13. Grasse L.W.N, **Stone S.A.**, Tata M.S. (2017). Localization of Moving Sound Sources in Virtual Reality Technology. Presented at the Undergraduate Research in Science Conference of Alberta. April 7-8; Calgary AB.

12. **Stone S.A.**, Tata M.S. (2017). Seeing Through Sound: A novel visual-to-auditory augmented reality device. Presented at the Canadian Spring Conference on Behaviour and Brain. February 23-25; Fernie BC.

11. Lee V., **Stone S.A.**, Tata M.S. (2017). What's in a cocktail party? Testing the mechanisms of auditory spatial resolution in multi-talker scenes. Presented at the Canadian Spring Conference on Behaviour and Brain. February 23-25; Fernie BC.

10. Grasse L., **Stone S.A.**, Tata M.S. (2016). The Effect of Head Movements on Auditory Front/Back Ambiguity in Virtual Reality Technology. Presented at the Prairie Perception, Action and Cognition Conference (P-PACT). November 5; Canmore AB.

9. Lee V., **Stone S.A.**, Tata M.S. (2016). How many guests makes for a good cocktail party? Presented at the Prairie Perception, Action and Cognition Conference (P-PACT). November 5; Canmore AB.

8. **Stone S.A.**, Delbrück T., Gonzalez C. L. R., Tata M.S. (2016). Prosthetic attention: A visual-to-auditory augmented reality system. Presented at the Prairie Perception, Action and Cognition Conference (P-PACT). November 5; Canmore AB.

7. **Stone S.A.**, Delbrück, T., Gonzalez C. L. R., Tata M.S. (2016). Computational model of motion sensitive cells using a neuromorphic artificial retina. Presented at the Banff Annual Seminar in Cognitive Science (BASICS). April 29-30; Banff AB.

6. **Stone S.A.**, Flindall J.W., Gonzalez C. L. R. (2015). The bias in our vision: Lower visual field advantages for grasping revealed through gaze analysis. Presented at the Cognitive Neuroscience Society Conference on Innovations in Mind and Brain Science. March 28-31; San Francisco, CA, USA.

5. **Stone S.A.**, Gonzalez C.L.R. (2015). Right visual field advantages for grasping, but only in the lower visual field. Presented at the Canadian Spring Conference on Behaviour and Brain. February 19-21; Fernie BC.

4. **Stone S.A.**, Cribben, I., Beal D.S. (2014). Functional Connectivity Analysis in Adults Who Stutter. Presented at the University of Alberta Research Experience presentation ceremony. August 20; Edmonton AB.

3. **Stone S.A.**, Gonzalez, C.L.R. (2014). The attention-hungry left space: An examination of eye gaze during a reach-to-grasp task. Presented at the Banff Annual Seminar in Cognitive Science (BASICS). May 2-4; Banff AB.

2. **Stone S.A.**, deBruin N., Gonzalez, C.L.R. (2014). Do you grasp where you look? Presented at the Canadian Spring Conference on Behaviour and Brain. February 20-22; Fernie BC.

1. MacLean J., **Stone S.A.**, Gonzalez C. L. R. (2013). EMG correlates of action observation. Presented at the Alberta Motor Control Association Annual Meeting (NeuroHike). September 27-29; Jasper AB.

AWARDS AND HONOURS

Mitacs Elevate Postdoctoral Fellowship ($120000)	May 2023–2025
Alberta Machine Intelligence Institute Travel Bursary ($500)	May 2022
Mitacs Accelerate Internship Award ($50000)	February 2021–December 2022
Mitacs Accelerate Internship Award ($30000)	February 2020–2021
Dept. of Psychology Best Graduate Student Research Paper Award ($500)	August 2019
President's Doctoral Prize of Distinction ($10000 (annual))	September 2019-2021
NSERC Postgraduate Scholarship-Doctoral ($42000)	September 2019-2021
Mitacs Accelerate International Internship Award ($45000; declined)	March 2019
Queen Elizabeth II Graduate Scholarship - Doctoral ($15000)	October 2018
University of Alberta Graduate Travel Award ($500)	October 2018
University of Alberta Skitch Travel Award ($250)	September 2018
University of Alberta Graduate Travel Award ($500)	November 2017
University of Alberta Department of Psychology Scholarship ($10000)	September 2017
University of Lethbridge Graduate Scholarship ($16000; declined)	July 2017
Government of Alberta Graduate Scholarship Award ($3000)	February 2017
Agility Neuroscience Business Plan Pitch Competition ($500)	April 2016
University of Lethbridge Graduate Travel Award ($500)	March 2015
University of Lethbridge Students Union Travel Award ($250)	March 2015
University of Alberta Research Experience Award ($6500)	May–August 2014
University of Lethbridge Students Union Travel Award ($500)	March 2014
University of Lethbridge Excellence & Leadership Qualities Scholarship ($2500)	January 2012
Don Henneberg Memorial Scholarship ($1500)	January 2012
Lufkin Academic Scholarship ($2000)	January 2012
KUDU Academic Excellence Scholarship ($750)	September 2011

GRANTS

University Hospital Foundation Medical Research Grant ($35000) January 2022–2023
Rapid Clinical Assessment of Vertigo Using Non-invasive Eye-tracking and Markerless Motion-tracking Techniques
Role: Investigator

Mitacs Accelerate Grant ($75000) – BioWare February 2021–December 2022
Developing Gaze and Movement Key Performance Indicators for User Experience Assessment
Role: Co-investigator

Mitacs Accelerate Grant ($45000) – BioWare February 2020–2021
Developing Gaze and Movement Key Performance Indicators for User Experience Assessment
Role: Co-investigator

ACADEMIC SERVICE

Conference Organization

Hidden Methods Conference (Virtual)–Lead organizer	April 2022
CAPnet-CPS CAN-ACN Satellite Symposium (Vancouver, BC)	May 2018
P-PACT (Perception Action Cognition) Conference (Canmore, AB)	November 2016
SfN Lethbridge Chapter Poster Symposium (Lethbridge, AB)	November 2016

Volunteer & Service

Dark Matters: The Science of Drugs, Presenter	October 2019
Telus World of Science NGSA Graduate Showcase, Presenter	March 2019
Science Fundamentals (SciFun), University of Alberta, Presenter	March 2019
Faculty of KSR Open House (University of Alberta), Presenter	Sept. 2018, Feb. 2019
Undergraduate Neuroscience Conference (University of Alberta), Judge	August 2018
Faculty of Medicine & Dentistry Festival of Health, Presenter	May 2018
Telus World of Science Graduate Student Showcase, Presenter	May 2018, 2019
Brain Awareness Week (University of Lethbridge), Setup	March 2016-17
Let's Talk Science (University of Lethbridge), Presenter	June 2014
Operation Minerva (University of Lethbridge), Presenter	April 2014

Ad Hoc Reviewer

IEEE Transactions on Human-Machine Systems (1)
Experimental Brain Research (2)
Frontiers in Psychology (1)
Frontiers in Neuroscience (1)
Journal of Cognitive Neuroscience (1)
PLOS ONE (1)
Journal of Advanced Research (1)
Trends in Cognitive Sciences (1)
Behavioural Brain Research (1)

Invited/Guest Lectures

Gaze and Movement Analysis (GaMA): Applications for rapid clinical assessment May 2021
Neuroscience Rounds, Division of Neurology, Faculty of Medicine, University of Alberta
Synchronizing Eye-Tracking and Motion Capture Data November 2018
PSYCO 104 (Introductory Psychology), University of Alberta

STUDENT SUPERVISION

Keeya Beausoleil (WISEST Student Summer 2019)
Jared Baker (publication)
Rob Olsen (publication)

TECHNICAL STRENGTHS & EXPERIENCE

Languages in order of familiarity	MATLAB, Python, C#, C++, Rust, LaTeX
Methods	eye-tracking, motion capture, reinforcement learning